

Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA5003
Course title (English)	Stochastic Process
Course title (Chinese)	随机过程
Units	3
Language of Instruction	English
Description (English)	This course introduces basic techniques for modelling and analyzing systems in the presence of uncertainty. It will cover Poisson processes, discrete and continuous Markov chains, martingales, Brownian motions, stochastic calculus and its applications in financial engineering.
Description (Chinese)	这门课程介绍了在不确定性情况下建模和分析系统的基本技术。课程将涵盖泊松过程、离散和连续马尔可夫链、布朗运动等基础知识。同时，还将展示涉及随机过程的各种应用示例。

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Prerequisites / Co-requisites

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course title.)

A. Prerequisites

University level linear algebra, calculus, probability theory, or MAT1040 and its equivalence.

B. Co-requisites

3. Learning Outcomes

Upon completing this course, the student will be able to:

- Understand fundamental concepts in stochastic models.
- Obtain knowledge about Markov chains, Poisson processes, martingales, and Brownian motion.
- Understand basic concepts of stochastic calculus
- Understand the basic principles of risk neutral pricing
- Acquire some basic techniques to use the tools of stochastic processes to establish proper models for financial problems and to perform quantitative analysis.

4. Course syllabus

- Discrete Markov chains
- Poisson processes and their applications
- Continuous-time Markov chains
- Martingale theory
- Brownian motion
- Introduction to stochastic calculus, Ito formula, and their applications in finance

5. Assessment Scheme

Component/ method	% weight
Assignments	20
Midterm Examination	40
Final Examination	40

6. Grade descriptor

General

Grade	Description
A	Demonstrate fully understanding of the course materials in ALL assignments, midterm and final exams
B	Demonstrate fully understanding of the course materials in MOST of the assignments, midterm and final exams
C	Demonstrate fully understanding of the course materials in SOME of the assignments, midterm and final exams
D	Demonstrate fully understanding of the course materials in LIMITED of the assignments, midterm and final exams
F	Demonstrate fully understanding of the course materials in VERY LIMITED of the assignments, midterm and final exams

7. Feedback for evaluation

- In-class Course and Teaching Evaluation
- Off-class feedback to instructor and/or teaching assistants Program review

8. Reading

Required Reading:

1. Introduction to Probability Models, ISBN-10. 0124079482, Sheldon M. Ross, 11, United States

9. Course components

Activity	Hours/week
Lectures	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Markov chains I: transition matrix, Chapman-Kolmogorov equations
2	Markov chains II: limiting probabilities and some applications
3	Poisson processes I: exponential distributions, Poisson process and its properties
4	Poisson processes II: time-inhomogeneous Poisson processes, compound Poisson process, mixed Poisson processes
5	No Class (National Day Holiday)
6	Continuous-time Markov chains I: Construction, Kolmogorov backward and forward equations, transition probability calculation
7	Review
8	Midterm Examination
9	Continuous-time Markov chains II: limiting probabilities
10	Martingale: definition and examples, stopping times, optional sampling theorem and its applications
11	Brownian motions I: scaled random walk and its limiting process, properties of BM, quadratic variations
12	Brownian motions II: first passage time, reflection principle, the maxima of BM, and the gambler's ruin problems.
13	Option Pricing
14	Review
15	Final Examination

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

Lecture: 100 student in one class (x 1)

12. Any other information

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

14. Version date

Version number	1
As of (date)	2025-07-24